Climate Risk Management 36 (2022) 100432

Contents lists available at ScienceDirect

Climate Risk Management

journal homepage: www.elsevier.com/locate/crm

Reviewing the nature and pitfalls of multilateral adaptation finance for small island developing states

Ellis Kalaidjian[a], Stacy-ann Robinson[b,*]

[a] Department of Marine Affairs, University of Rhode Island, Greenhouse Road, Kingston, RI 02881, United States
[b] Environmental Studies Program, Colby College, Mayflower Hill Drive, Waterville, ME 04901, United States

ARTICLE INFO

Keywords:
Adaptation
Climate change
Finance
Governance
Small Island developing states (SIDS)

ABSTRACT

Small island developing states (SIDS) are recognized as being among the most vulnerable countries to climate change—many are highly exposed and sensitive to, *inter alia*, sea-level rise and increased intensity storms, yet often lack the human and financial resources needed for successful and/or effective adaptation. The international community has responded to SIDS' precarious situation, in part, by establishing a suite of multilateral funding mechanisms to help finance costly adaptation projects and programs. However, to date, there has been a mismatch between the level of international climate finance mobilized and the scale of adaptation that is needed in SIDS. This paper reviews the relevant academic and policy literatures to provide a system-wide diagnosis of the institutional and conceptual problems plaguing adaptation-specific multilateral climate funds, which are often addressed in isolation. We show that, following the establishment of the Global Environment Facility, many multilateral climate funds have been rife with administrative issues, creating barriers to access for particularly vulnerable countries. The funding that SIDS *do* receive is often insufficient to meet the costs of adaptation, and funding allocations are demonstrably skewed amongst SIDS *and* amongst types of adaptation projects and programs funded. In concluding, we propose reorienting multilateral adaptation finance towards resilience as a means of improving the governance of international adaptation finance and increasing access for SIDS and other particularly vulnerable countries.

1. Introduction

Small island developing states (SIDS) comprise a heterogeneous group of 58 countries scattered across three main geographic regions—the (1) Caribbean, (2) Pacific, and (3) Atlantic, Indian Oceans, Mediterranean and South China Seas (also called the AIMS region) (Robinson, 2017). Thirty-eight of these countries are United Nations Member States. The 'small island developing states' categorization, mainstreamed in the outcomes of the 1992 Rio de Janeiro Earth Summit, fosters awareness of the shared challenges that these countries face in managing climate change—e.g., "… limited resources, geographic dispersion and isolation from markets …" (United Nations, 1992, p. 193). Yet, while collectively grouped in this way, SIDS differ in their geography, history, cultures, and economic development. For instance, not all SIDS are small (e.g., Papua New Guinea in the Pacific is the 54th largest country), or islands (e.g., Guyana is a low-lying coastal country in South America, but considered part of the Caribbean), or developing (e.g., Singapore in the AIMS region is a high-income economy), or independent States (e.g., the British and US Virgin Islands are overseas

* Corresponding author.
E-mail address: stacy-ann.robinson@colby.edu (S.-a. Robinson).

https://doi.org/10.1016/j.crm.2022.100432
Received 2 December 2021; Received in revised form 23 March 2022; Accepted 20 April 2022
Available online 27 April 2022
2212-0963/

dependencies in the Caribbean) (Robinson, 2018a). Many SIDS are middle-income countries, but often with small economies and considerable variation in gross national income. Eight are currently co-classified as least developing countries (LDCs) because of their low national incomes, high exposure to economic and environmental shocks, and low levels of human assets (Robinson, 2020a).

SIDS are, therefore, recognized as a "special case both for environment and development" in that they have "all the environmental problems and challenges of the coastal zone concentrated in [limited land areas]" (United Nations, 1992, p. 193). Chapter 15 of Working Group II's contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, considers all small islands to be particularly vulnerable to the impacts of sea-level rise and increasingly severe tropical and extratropical cyclones (Mycoo et al., 2022). Increased sea surface temperatures, ocean acidification, and changing rainfall patterns associated with climate change also jeopardize the ecologically fragile marine ecosystems on which they rely (Mycoo et al., 2022). Because of their high degree of climate change exposure and sensitivity, SIDS face relatively high adaptation costs (Robinson & Dornan, 2017). For instance, the costs of shoreline protection structures to combat sea-level rise and storm surge in SIDS are substantially greater, in terms of both per capita and percentage of gross domestic product (GDP), than the costs of similar infrastructure in larger territories with larger populations and higher levels of gross national income per capita (Nunn et al., 2021; Nurse et al., 2014; Piggott-McKellar et al., 2020). The macroeconomic impacts of climate change on SIDS' economies cannot be understated either. Bueno et al. (2008) estimate that Caribbean SIDS could face annual climate-related losses in excess of US$22 billion by 2050. A more recent study by Cabezon et al. (2019) demonstrates that natural hazards, particularly those exacerbated by climate change, collectively contribute to higher revenue volatilities, damage growth prospects, lower potential growth rates, and often deteriorate their fiscal capacities. The authors' cross-country analyses of hazard impacts on SIDS' economies further suggest that, in the short-term, natural hazards causing damage and losses equal to 1% of GDP produce an average 0.7% drop in GDP and deterioration in fiscal balance of 0.5% GDP (in the year of the disaster); in the long-term, disasters producing damage and losses equal to 1% of GDP can decrease GDP trend growth by up to 0.7 percentage points over 10 years (Cabezon et al., 2019).

Given that continued climate change is projected to increase the intensity of natural hazards such as tropical and extratropical cyclones, the ability of many SIDS' economies to rebound post-disaster may be hindered by back-to-back storm impacts. Further complicating SIDS' situation, their capacities to adapt to such changes—another key component of their vulnerability (Smit & Wandel, 2006)—are significantly constrained by numerous economic and sociopolitical barriers (Robinson, 2018b). For example, many SIDS have narrow resource bases, depriving them of the benefits of large economies, as well as small domestic markets and large reliance on few external and remote markets (Nurse et al., 2014; Robinson, 2018b). Many SIDS are also physically isolated from export markets and import resources, and have weak(er) governance systems that are not well-suited to manage the complexity and uncertainty of climate change. Additionally, there is limited private sector support for SIDS' adaptation efforts, resulting in a proportionately large dependence on public resources (Cabezon et al., 2019; Nunn et al., 2013; Robinson, 2019).

In light of SIDS' high exposure to climate change and limited resources to adapt, the international community has been called on to assist them in "[designing] and [implementing] rational response strategies to address the environmental, social, and economic impacts of climate change and sea-level rise …" (United Nations, 1992, p. 194). SIDS are also particularly reliant on international financial assistance to combat the impacts of climate change at the national and sub-national levels. Academic interest in SIDS' experience with international adaptation financing is increasing. However, in recent years, much of the work published in the environmental studies/science literature has focused on empirical analyses of funding commitments or flows, primarily using datasets produced by bilateral and multilateral donors such as the Organisation for Economic Co-operation and Development (OECD)[1]. A much smaller part of the literature focuses on the governance and politics of multilateral climate adaptation finance in relation to the circumstances of SIDS, though studies such as Roberts et al. (2021) focus broadly on the experience of developing countries.

This desk-based review helps bridge two bodies of work. It conducts a narrative review of the climate finance landscape for SIDS and looks to the published academic and policy literatures to conduct a systems-wide diagnosis of the institutional and conceptual problems plaguing adaptation-specific multilateral climate funds, which are often addressed in isolation. To achieve this, the remainder of this paper substantively covers the principles, architecture, and bureaucracy of multilateral climate finance (Section 2), what the SIDS context reveals about the *nature* of multilateral adaptation finance (Section 3), what the SIDS context reveals about the *pitfalls* of multilateral adaptation finance (Section 4), and a proposal for reorienting multilateral adaptation finance towards resilience (Section 5). In focusing on multilateral funds, however, we only cover bilateral flows in passing. These flows, despite their increasingly important role in the climate finance landscape (e.g., see Buchner et al., 2019), largely fall outside the scope of this paper.

2. Multilateral climate finance: Principles, architecture, and bureaucracy

Over the course of the 20th century, the international community has created multilateral finance organizations to facilitate economic and social development in developing countries (Nelson, 2012). Notable examples include the International Monetary Fund and the World Bank, which provide a mixture of concessional and non-concessional financial assistance to low- and middle-income countries (Nelson, 2012). Climate change poses additional challenges to developing countries' social and economic growth prospects, and building climate change considerations into development agendas incurs costs beyond normal growth trajectories (Lattanzio, 2014). Hence, over the last several decades, the international community has established, *inter alia*, multilateral climate funds (MCFs) to channel "climate finance" to State Parties to the United Nations Framework Convention on Climate Change (UNFCCC or the

[1] See, for example, Robinson and Dornan (2017) and Weiler et al. (2018).

Convention). While interpretations of climate finance are variable (Roberts et al., 2021; Robinson, 2020a,b; Weikmans et al., 2020), it is generally understood as finance that specifically aims to either mitigate greenhouse gas emissions (hereon referred to as 'mitigation finance') or "[reduce the] vulnerability of, and [increase] the resilience of, human and ecological systems to negative climate change impacts" (hereon referred to as 'adaptation finance') (UNFCCC, 2014, p. 5).

2.1. The principles of climate finance

The UNFCCC sets forth a normative framework instructing how, to whom (or to what causes), and by whom climate finance should be administered through the MCFs operating under the Convention. Articles 3, 4, and 11 of the Convention, which identify principles, commitments, and an operational mechanism, respectively, guide finance. Article 3(1), for example, obliges developed country Parties to take climate actions, including financial assistance, on "the basis of *equity* and in accordance with their *common but differentiated responsibilities and respective capabilities*" (UNFCCC, 1992, emphasis added). Further, the administration of climate finance must ensure "an *equitable and balanced representation of all Parties* within a *transparent* system of governance" (UNFCCC, 1992, emphasis added). Considering the significant exposures and limited adaptive capacities of many developing countries, SIDS included, the Convention outlines specific provisions for adaptation finance found under subsections of Article 4. Under Article 4(3), for example and in accordance with the principle of common but differentiated responsibilities and respective capabilities, developed countries must provide "*new and additional* financial resources [...] needed by developing country Parties to meet the agreed full incremental costs of implementing [adaptation] measures" (UNFCCC, 1992, emphasis added). These commitments "shall take into account the need for *adequacy and predictability* in the flow of funds and *the importance of appropriate burden sharing* among the developed country Parties" (UNFCCC, 1992, emphases added). Importantly, the Convention also outlines special funding provisions for "developing country Parties that are *particularly vulnerable* to the adverse effects of climate change" (Article 4(8), emphasis added) which includes "small island countries" (Article 4(8)(a)) and "countries with low-lying coastal areas" (Article 4(8)(b)), i.e., SIDS. Later agreements such as the 2010 Cancun Agreements reaffirmed these imperatives and enforced additional provisions, including the need for climate finance to be "*balanced"* for adaptation measures, with an emphasis on the needs of SIDS and other particularly vulnerable countries. The 2015 Paris Agreement further reaffirms that the "[p]rovision of resources should also aim to achieve a balance between adaptation and mitigation ..." (UNFCCC, 2016, online), requiring developed country Parties "... to submit indicative information on future support every two years, including projected levels of public finance" (UNFCCC, 2016, online), in addition to reporting on finance already provided. The Agreement also establishes that the financial mechanism of the Convention—i.e., the Global Environment Facility (GEF)—and the Green Climate Fund (GCF), would help Member States meet their obligations under Paris (UNFCCC, 2016).

2.2. Climate finance negotiations in the UNFCCC

Since the UNFCCC entered into force in 1994, there have been deep divisions with respect to climate finance within the Conference of the Parties (COP), the supreme decision-making body of the Convention. These divisions broadly relate to defining 'climate finance', accountability and transparency, and failures to create corresponding modalities, influencing decisions on the nature of long-term finance (Munira et al., 2021). Additional points of disagreement have included reporting format, private sector involvement, public intervention, and increased involvement of the OECD and multilateral development banks (MDBs) such as the World Bank and the Inter-American Development Bank[2] (Munira et al., 2021).

Divisions within the COP complicate the progress of climate finance negotiations. Dividedness over the role of the GEF led to the collapse of negotiations at COP6 in The Hague in 2000, for example. There, developing countries expressed concern about (1) the types of adaptation activities the GEF should fund and (2) the modalities for such funding, (3) whether the GEF should fund capacity building for disaster preparedness and disaster management (e.g. early warning systems for extreme weather events), and (4) whether the GEF should be the only channel for funding in certain areas, including technology transfer (Dessai, 2001). Similar disagreements resurfaced at COP15 in Copenhagen in 2009—developing countries pushed for a new operating entity but faced resistance from developed countries that were in support of the existing financial arrangements with the GEF whereby decision-making in the Council was by consensus (Liverman & Billett, 2010). As with COP6, divisions on public versus private financing, additionality, adequacy, and predictability of financing at COP15 also stymied negotiations over climate finance. In the end, countries only 'took note' of the Copenhagen Accord, the key provisions of which included (1) scaled up, new and additional, predictable and adequate funding, (2) improved access for developing countries, (3) US$30 billion in new and additional financing for 2010–2012 ("fast start"), (4) US$100 billion per year by 2020, (5) balanced allocation between mitigation and adaptation objectives, (6) 'particularly vulnerable countries' being prioritized for funding, (7) equal representation in governance structures, and (8) a significant portion of funding to flow through the Copenhagen 'Green Climate Fund' (Rajamani, 2010).

[2] Other examples of MDBs include the African Development Bank, the Asian Development Bank, the Asian Infrastructure Investment Bank, the European Bank for Reconstruction and Development, the European Investment Bank, and the Islamic Development Bank.

2.3. The global climate finance architecture

Today, climate finance flows through a complex network of over 500 bilateral, regional, and multilateral initiatives and channels, and is offered in forms ranging from grants and concessional loans, to guarantees and private equity (Bracking & Leffel, 2021; Khan et al., 2020; Watson & Schalatek, 2021). There are over 20 multilateral funds and initiatives, including the GEF and the GCF, which are the two best-known UNFCCC financial mechanisms. Article 12.8 of the 1997 Kyoto Protocol to the Convention called for a "share of proceeds" from certified project activities under its Clean Development Mechanism to be used to cover administrative expenses. This led to the creation of the Adaptation Fund, which is financed largely by public and private sector donors (UNFCCC, 2021a). MCFs outside the umbrella of the UNFCCC include the Pilot Program for Climate Resilience and the Adaptation for Smallholder Agriculture Programme. The Pilot Program for Climate Resilience is a targeted program of the Strategic Climate Fund, which is one of two funds under the Climate Investment Funds framework administered by the World Bank. In addition to these multilateral funds and initiatives, there are eight or more major bilateral funds and initiatives, including the Global Climate Change Initiative (United States) and the International Climate Fund (United Kingdom). A growing number of recipient countries are also establishing national climate funds that are either entirely domestically funded or that receive monies from multiple contributor countries to coordinate and align contributor interests with national priorities (Watson & Schalatek, 2021). Among the 15 or so national and regional funds is the Maldives Climate Change Trust Fund, one of the few among SIDS. There are also around 30 major implementing agencies and institutions, including the UN Development Programme, UN Environment Programme, regional development banks such as the African Development Bank, and bilateral agencies such as the Norwegian Agency for Development Cooperation and the United States Agency for International Development. Fig. 1 below provides an overview of some of the main public multilateral climate finance mechanisms. It is important to note that these mechanisms are only part of a very complex network of mechanisms, and that each mechanism has its own governance structure, modality and objectives, which gives rise to incredible variance across the network (Bracking & Leffel, 2021; Watson & Schalatek, 2021).

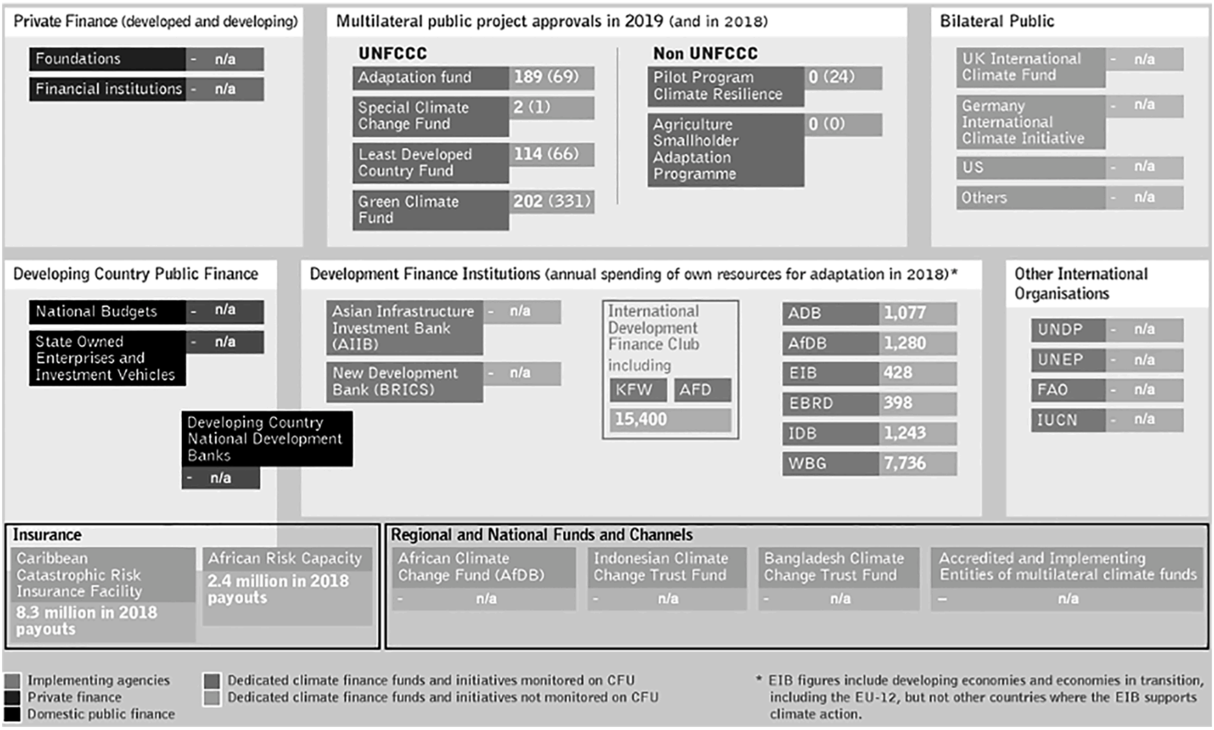

Fig. 1. Overview of some of the main public multilateral climate finance mechanisms, (Modified from Watson & Schalatek, 2021, p. 2, with permission).

2.4. The bureaucracy of access

The process for accessing climate finance through any channel is equally as complex as the architecture described above. In sum, developed countries contribute to an MCF, which, in the case of the MCFs under the Convention, are overseen by developing and developed members of the COP, in accordance with the Convention's provision of equitable representation. Eligible developing country applicants, determined differently by each MCF, then submit funding proposals through an accredited implementing entity, which may be a (sub-) national, regional, national, or multilateral organization (Omari-Motsumi et al., 2019). Accredited entities are selected and nominated by a government official from the applicant country who serves as a point of contact for the relevant MCF. These National Designated Authorities are responsible for administrative tasks such as programming and delivery (Omari-Motsumi et al., 2019). In the case of the GCF, unlike the MDBs, countries do not require membership to access funds. Instead, a developing country that is a UNFCCC Party appoints a National Designated Authority that acts as the interface between its government and the GCF. The GCF then relies on a network of International Accredited Entities—e.g., the UN Development and Environment Programmes, and MDBs such as the Inter-American Development Bank—and Direct Access Accredited Entities—e.g., national and regional entities such as the Fiji and Caribbean Development Banks—to implement approved projects. Good implementing entity candidates often have robust track records of implementing projects and effective and documented processes to reduce fiduciary, environmental, and social risk (WRI, 2015). Project management experience and a track record of implementation success thereby affect a country's ability to attract future adaptation financing (following Ameli et al., 2020). This is no different from clean energy investment decisions, for example (e.g., see Nelson & Pierpont, 2013). This then opens up a role for, e.g. state investment banks to create trust for projects and to help organizations and projects gain a track record (Geddes et al., 2018).

Entity accreditation processes vary between MCFs, but MCFs often contain sets of fiduciary standards that National Designated Authorities must consider in their implementing entity appointment decisions, such as legal status, financial and management integrity, institutional capacity, and transparency, self-investigation and anti-corruption (Adaptation Fund, 2014). Following entity accreditation and proposal submission, proposals are reviewed by the distinct organs of the MCFs—in the case of the GEF, proposals are evaluated by the GEF Council, its executive organ, the GEF Secretariat, and its Scientific and Technical Advisory Panel. Input into funding allocation decisions may also come from representatives from civil society organizations and other stakeholders during panel meetings (Omari-Motsumi et al., 2019), consistent with the Convention's obligations of transparent governance. Upon project approval, funds from the MCF flow through implementing entities to executing agencies, which are then responsible for all execution functions, from project conceptualization to implementation, and the local level delivery of funds. The general process through which potential beneficiary countries, including SIDS, may access MCF funding is simplified and illustrated in Fig. 2 below.

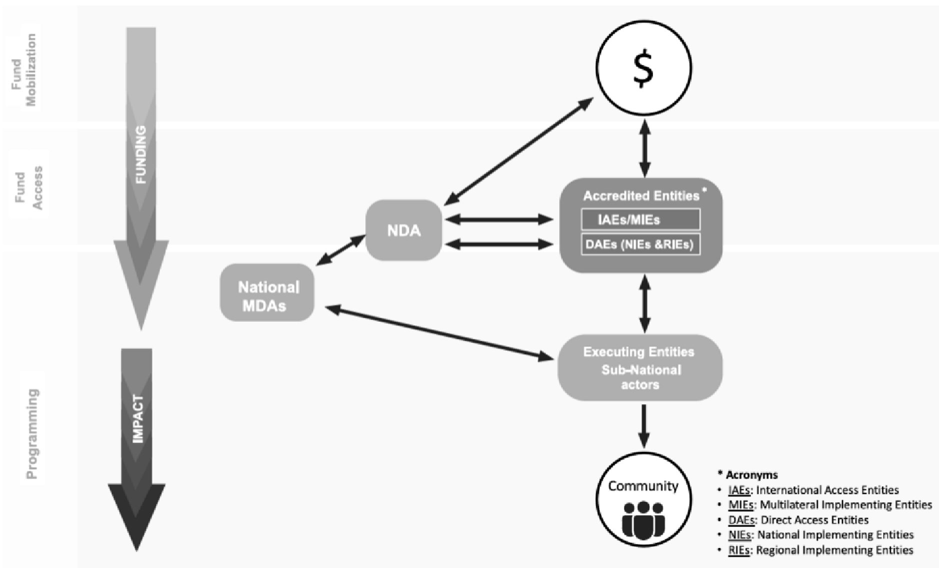

Fig. 2. A simplified illustration of the flow of climate finance from MCFs to beneficiary countries, (Modified from Omari-Motsumi et al., 2019, p. 7, with permission).

Following the establishment of the first MCF, the GEF, the application process for MCF funding was criticized for being too bureaucratically cumbersome and expensive[3] for SIDS, LDCs, and other developing countries with limited human resources (Robinson & Dornan, 2017). Early on, implementing entity accreditation was contentious because multilateral institutions were often nominated by National Designated Authorities. This required subnational developing country leaders to go through an intermediary to access the funding they were awarded, which frequently led to issues associated with inefficient fund distribution at the local level, where adaptation action must take place (Omari-Motsumi et al., 2019). However, a pivotal development in the climate finance architecture was brought about by the establishment of the Adaptation Fund in 2001. The Adaptation Fund "finance[s] concrete adaptation projects and programmes in developing country Parties to the [UNFCCC's] Kyoto Protocol that are particularly vulnerable to the adverse effects of climate change" (UNFCCC, 2021a, online). The establishment of the Adaptation Fund was also meant to facilitate direct access to allow accredited national, regional, and sub-regional entities from developing countries to directly access financing, without having to go through an international intermediary institution. The progressive design of the Adaptation Fund informed the governance arrangement of the GCF, which was established in 2010 and which was fully operationalized in 2014 "as a dedicated financing vehicle for developing countries within the global climate architecture, serving as the financial mechanism of the UNFCCC [and later the Paris Agreement]" (GCF, 2021, online). Despite the progressive design, the GCF has been critiqued for, among other things, not paying enough attention to critical result areas (Fonta et al., 2018), reflecting divides over financialization and country ownership (Bertilsson & Thörn, 2021), and not embracing more innovative forms of climate finance such as blockchain (Schulz & Feist, 2021).

The administrative and operational procedures of the GCF, in particular, have also been censured. Chen (2018), for example, highlights the Fund's inability to integrate the interests of different countries and questions the sustainability of its resources, its investment direction, and the capability of the Secretariat. The GCF's investment direction is guided by a six-component framework: (1) impact potential, (2) paradigm shift potential, (3) sustainable development potential, (4) recipient need, (5) country ownership, and (6) efficiency and effectiveness. Anecdotal evidence suggests that many SIDS do not routinely develop proposals in alignment with these criteria, including requirements for gender mainstreaming. With respect to Component 2, Kasdan et al. (2021, p. 427) express concern over the "increasing emphasis on transformational adaptation" across Funds, but especially in the GCF, all while "a clear understanding of whether transformational change is achievable, feasible, and desirable under all conditions has not yet emerged". Bertilsson and Thörn (2021, p. 437), in their GCF critique, show "how the concepts of transformational change and paradigm shift have been mobilized as discursive resources in order to control the project proposal process, while still describing it as country ownership and responsiveness to the needs of recipient countries". This, they argue, serves to "legitimize a top-down financialization of recipient countries" (Bertilsson & Thörn, 2021, p. 437). To help address this concern in SIDS and as an example, the United Nations Office for Project Services has helped develop a national infrastructure financing strategy for St. Lucia (see Adeoti et al., 2021). The strategy "builds on the National Infrastructure Assessment, which provides a robust pipeline of projects to meet long-term infrastructure needs, in line with national development objectives, as well as the Sustainable Development Goals and the Paris Agreement" (Adeoti et al., 2021, p. 4). Recognizing that accessing adequate financing for project implementation is still a challenge for the country, the strategy demonstrates how "gender considerations [can] highlight the transformative power of infrastructure to improve the lives of women and girls" (Adeoti et al., 2021, p. 20).

3. The *nature* of multilateral adaptation finance: Actors and actions

Increasing emphasis on developing countries' adaptation needs has resulted in MCFs becoming more adaptation focused (Watson et al., 2019; Watson & Schalatek, 2020b). Between 2003 and 2019, there were 12 MCFs active in SIDS (see Table 1 below), and seven specifically funding adaptation projects or funding both adaptation and mitigation projects (hereon referred to as multilateral adaptation funds or MAFs). The primary MAFs supporting adaptation in SIDS include the GEF and its two specialized funds—the LDC Fund and the Special Climate Change Fund—the Adaptation Fund, and the largest and newest fund, the GCF, all of which operate under the Convention. The LDC Fund was established in 2001 to support the LDC work programme under the UNFCCC and included the preparation and implementation of national adaptation programmes of action. As of October 2019, 51 LDCs and former LDCs had accessed a total of US$1.4 billion for the preparation and implementation of national adaptation programmes of action, the national adaptation plan process, and elements of the LDC work programme (United Nations, 2021a). The Special Climate Change Fund, which was also established in 2001, finances adaptation and technology transfer (both mitigation and adaptation). Graduated LDCs and other developing countries, including SIDS, have access to the Special Climate Change Fund for the purpose of elaborating and implementing their national adaptation plan processes. Two additional MAFs, the Pilot Program for Climate Resilience, and the Adaptation for Smallholder Agriculture Programme, operate separately under the World-Bank-managed Climate Investments Funds, which channel adaptation finance through regional development banks that work with national governments, private sector project sponsors, financial institutions, development partners, and other stakeholders (World Bank, 2020). Additionally, the European Commission-launched Global Climate Change Alliance provides support to developing countries most vulnerable to the impacts of climate change, particularly SIDS and LDCs, by strengthening dialogue and cooperation on climate change and offering technical support.

Over the last five years, adaptation finance mobilized through MAFs has increased in SIDS (Watson & Schalatek, 2020a). Of the seven MAFs, the GCF has become the largest contributor since 2015, approving a cumulative US$646 million for SIDS. A recent independent evaluation showed that this amount increased to US$818 million across 29 projects, representing "a reasonable proportion

[3] Among the major funds, fees account for between 1% and 9% of total fund value, ranging from US$65,000 to US$1.2 million per project (Watson et al., 2019). SIDS' frustration with intermediary fees is addressed in Robinson and Dornan (2017), for example.

Table 1
Multilateral adaptation funds supporting SIDS (2003–2019).

Funds and Initiatives	Theme	Amount approved (USD Millions)	Number of projects approved
1. Green Climate Fund	Multiple foci	645.7	19
2. Pilot Programme for Climate and Resilience	Adaptation	226.5	18
3. Least Developed Countries Fund	Adaptation	194.2	51
4. Global Environment Facility	Multiple foci	172.1	76
5. Adaptation Fund	Adaptation	134.6	33
6. Global Climate Change Alliance	Adaptation	99.5	20
7. Scaling-Up Renewable Energy Program for Low Income Countries	Mitigation	78.2	10
8. Forest Carbon Partnership Facility	Mitigation	67.4	14
9. Clean Technology Fund	Mitigation	56	4
10. Special Climate Change Fund	Adaptation	40.9	6
11. UN REDD Programme	Mitigation	6.9	2
12. Adaptation for Smallholder Agriculture Programme	Adaptation	5.1	2

(Source: Watson & Schalatek, 2020b, p. 2).

of total approved finance, in consideration of per capita representation" (GCF, 2020, p. 3). This suggests that the international community has achieved some degree of progress in leveraging the GCF as the primary mechanism in meeting the 2010 Cancun Agreements' target of balanced adaptation finance. The Climate Investments Funds have also played a significant role in financing climate action in SIDS, as evidenced by the US$226 million allocated through the Pilot Program for Climate Resilience, which is followed by the LDC Fund, which has approved US$194 million. Another optimistic trend is that a bulk of the funding (87%) that SIDS receive from MAFs (and other MCFs) is grant based, with concessional loans and guarantees constituting a much smaller proportion of the total (13%)—important due to the fact that many SIDS are heavily in debt and in dire need of grants rather than loans (Bourne et al., 2015). However, as we will show later, the grant/loan split varies by SIDS region with direct budget support being rare.

Co-financing has also risen in importance—the level of a project's co-financing can impact its selection, design, scale, and scope (Cui et al., 2020). Co-financing includes additional project resources in the form of grants, loans, guarantees, cash, and in-kind support from multilateral and bilateral sources, governments, private sector, civil society organizations, and the project beneficiaries themselves (Yu & Miller, 2011). Since its establishment in 1991, the GEF has provided US$16.55 billion in grants to 4,574 projects in 163 developing countries and cumulatively leveraged US$98.26 billion in capital (Cui et al., 2020). Its co-financing ratio increased from 2.37 in the first replenishment of resources (GEF-1) in 1994–1998, to 7.32 in the fifth replenishment of resources (GEF-5) in 2010–1014, though the average ratio between 1991 and 2018 was 5.94 (Cui et al., 2020). This means that, for every US$1 in grant funding awarded, US$5.94 in other capital would need to be leveraged. Therefore, every project needs to maximize the scale of its co-financing (Yu & Miller, 2011), though emerging economies are able to leverage more co-financing than lower income countries (Cui et al., 2020), including LDC-SIDS.

In their assessment of World Bank and Asian Development Bank project performance, Bulman et al. (2017, p. 335) concluded that "country-level macro variables, GDP growth, and the policy environment are significantly positively correlated with project outcomes". Therefore, country-level and project-level characteristics play a key role in funders' decision-making (Bulman et al., 2017), making it difficult for SIDS with low incomes and low GDP growth to access financing from these multilaterals. A 2021 study by the International Institute for Environment and Development compared the reported adaptation costs in 22 LDCs and countries' borrowing space—the difference between a country's sustainable debt-to-GDP threshold and its debt-to-GDP ratio. Total estimated adaptation costs across the 22 countries, including three LDC-SIDS (Comoros, Haiti, Solomon Islands), was almost US$212 billion (IIED, 2021). It is likely that countries could afford around 20% of this cost through sovereign debt before tipping into debt distress, which is likely to increase given the compounding nature of climate impacts. This suggests that LDCs are unable to borrow upward of 80% of the money needed to support adaptation efforts (IIED, 2021). To help remedy this dilemma, other SIDS such as Belize have sought Commonwealth support to explore the possibility of a debt-for-climate swap, as approximately 7% of its GDP (US$123 million) is being lost to climate-amplified events like extreme winds and floods (Commonwealth Secretariat, 2021). The additional fallout from COVID-19 has pushed the country's debt-to-GDP ratio to 130%, making access to climate readiness and support funds from such agencies as the GEF, GCF, and the Adaptation Fund even more difficult (Commonwealth Secretariat, 2021).

3.1. Sources and flows of adaptation finance are highly skewed across countries

A recent study by Garschagen and Doshi (2022), focusing on adaptation flows from the GCF since 2015, shows that the Fund's allocations are supporting the country groups that it aims to prioritize—African countries, LDCs and SIDS. Within these groups, African LDCs received 18% of total adaptation financing, while other African countries, and non-LDC and non-African SIDS, received 14% and 10%, respectively (Garschagen & Doshi, 2022). At the same time, countries belonging to the highest climate vulnerability quartile (57%), yet characterized by weak governance and fragile state-bureaucracies, have missed out and not been able to access project funding (Garschagen & Doshi, 2022). These countries include SIDS such as Guinea-Bissau, Federated States of Micronesia, Haiti, and Papua New Guinea. Additionally, 48% of the financing went to those countries with strong institutional capacity and topping the world rankings of government effectiveness maintained by the World Bank. High-ranking SIDS such as Antigua and Barbuda have had success in getting their national entities accredited and consequently accessing GCF financing (Garschagen & Doshi, 2022).

On the contrary, some of the most recent studies of OECD adaptation commitments to SIDS reveal that (1) a handful of SIDS receive the most funds, (2) bilateral donors provide the bulk of financing, and (3) countries with similar levels of vulnerability receive vastly different amounts (Canales et al., 2017; Donner et al., 2016; Robinson & Dornan, 2017). For instance, in the AIMS region, flows to two countries—Mauritius and Cabo Verde—accounted for 83% of total climate finance to the region; 90% of adaptation finance was provided by bilateral donors, primarily France and Japan (Canales et al., 2017). AIMS SIDS that are also classified as LDCs—Comoros, Guinea-Bissau, and São Tomé and Principe—received the least funds (Canales et al., 2017). In the Caribbean, five SIDS—Dominican Republic, Haiti, Guyana, Jamaica, and Cuba—accounted for 88% of total adaptation commitments, the bulk of which, again, came from France (Robinson, 2018). For Pacific SIDS, Australia was the most significant bilateral adaptation donor, providing 54% of OECD finance to the region between 2008 and 2012, and 18% across all SIDS between 2010 and 2014 (Donner et al., 2016; Robinson & Dornan, 2017). While having similarly low elevations and coralline compositions, and faced with similar physical limits to adaptation, countries such as Kiribati, the Marshall Islands, and the Federated States of Micronesia, received US$4.5 million, US$0.6 million, and US$0.8 million in explicit adaptation finance, respectively (Donner et al., 2016).

Despite observed inter-SIDS funding discrepancies, a country's classification as a SIDS remains a predictor of the amount of adaptation financing it can expect to receive, though estimates of total flows of adaptation finance into SIDS vary based on data source, time period, and definitions of 'adaptation' and 'adaptation finance', among other factors (Robinson & Dornan, 2017). In 2017, eight MDBs, including the Inter-American Development Bank and World Bank, financed adaptation to the tune of US$217 million in 37 SIDS, including Haiti, Papua New Guinea, and Jamaica, which received the largest amounts (European Bank for Reconstruction and Development et al., 2017). This represented 58% of total climate finance to these countries (European Bank for Reconstruction and Development et al., 2017). Robinson and Dornan (2017), who used data from the OECD, reported total adaptation finance at roughly US$2 billion across 50 SIDS between 2010 and 2014. The top ten per capita adaptation finance recipients in Betzold and Weiler (2017) are all SIDS. Niue, for example, received US$12,707 per capita, despite only having roughly 1,500 inhabitants (Betzold & Weiler, 2017). Other determinants of flows of finance into SIDS for adaptation include population, per capita income, governance quality, and vulnerability, but depend on how adaptation is conceptualized and measured (Robinson & Dornan, 2017).

3.2. *Adaptation finance is channeled towards 'general environmental protection'*

Though the results of Betzold and Weiler (2017) and Weiler et al. (2018) showed that donors 'privileged' vulnerable countries such as SIDS and those that are more physically exposed, adaptation finance commitments from the OECD to SIDS primarily targeted the water and sanitation, and energy sectors, and 'general environmental protection'. Around US$473 million was committed to Caribbean SIDS for 'general environmental protection', with less emphasis on those sectors such as education and health that are critical for resilience and sustainability; US$441 million was committed to Pacific SIDS for creating 'enabling enviornments' and mainstreaming adaptation (Atteridge & Canales, 2017; Atteridge et al., 2017). The tagging of adaptation finance for 'general environmental protection' is notable because of the tension in the literature about whether adaptation is development and whether adaptation finance can be differentiated from development finance, which would target economic, environmental and social programming. The epistemic ambiguities of "new and additional" and "adaptation" have received substantial criticism, largely in the context of bilateral and multilateral adaptation-specific official development assistance (Ayers & Abeysinghe, 2013). Scholars question whether aid constitutes finance that is 'new and additional' to development finance and how adaptation is different from development (Ayers & Abeysinghe, 2013; Hall, 2017), as adaptation actions are typically situated within an ongoing effort to achieve sustainable development at the national level (Robinson, 2019). It is, therefore, difficult to identify and isolate the additional cost and contribution of the adaptation project or program in order to merit the award of the additional funding (Brown & Kaur, 2009).

This difficulty also has implications for the traceability of adaptation finance, which is complicated by the fact that many MCFs and other multilateral organizations apply their own criteria to their definitions of adaptation, thereby violating the climate change regime's objective of transparency (Hall, 2017; Weikmans et al., 2019). For example, MDBs that fund adaptation projects across Member States require both an adaptation intent, the set-out of the climate vulnerability context of the project, and the articulation of a clear link between the context of climate vulnerability and the project (AfDB, 2014). More recent efforts to harmonize principles of adaptation finance across a network of MDBs and the International Development Finance Club, paint a broader definition of adaptation as activities that "[reduce] the vulnerability of people or natural systems to the impacts of climate change and risks related to climate variability by maintaining or increasing adaptive capacity and resilience" (World Bank, 2015, p. 3) The GEF, by contrast, has one of the narrowest conceptualizations of adaptation—it requires adaptation proposals to demonstrate additional costs imposed by climate change on top of an existing development baseline and will only fund the additional costs of climate change (GEF, 2011). This has been an enduring critique of the GEF, especially because it is one of the financial mechanisms of the UNFCCC (see Cui et al., 2020).

4. The *pitfalls* of multilateral adaptation finance: Equity-oriented norms and allocation criteria

There are growing concerns among policy- and decision-makers in SIDS about the current levels and forms of adaptation finance, and about countries' experience with accessing it (Robinson & Dornan, 2017). Though climate finance has been flowing, there is still neither an official definition of climate finance, nor mutually agreed accounting modalities (Roberts et al., 2021; Weikmans et al., 2020). This is in spite of the Paris Agreement 'rulebook' being adopted at COP24 in Katowice in 2018 and setting out the rules for climate finance accounting (Khan et al., 2020; Weikmans & Roberts, 2019; Weikmans et al., 2019), and in spite of COP26 in Glasgow in 2021 deciding to "initiate the deliberations on setting a new collective quantified goal and to conduct the deliberations in an open, inclusive and transparent manner, ensuring participatory representativeness" (UNFCCC, 2021b, online). These issues make the processes of coordinating, tracking, and accessing climate finance difficult. They further exacerbate the continuing proliferation of financial mechanisms (Pickering et al., 2017; Watson & Schalatek, 2021). In turn, the vastness of the climate finance landscape leads to many overlaps involving high transaction costs (Brunner & Enting, 2014; Giglio et al., 2021), which generate frustrations both for the contributors who are duplicating efforts and for the recipients who must complete tedious paperwork in order to access these funds (Robinson & Dornan, 2017). And while the transparency of climate finance programmed through multilateral initiatives is increasing, with evidence that "norms related to transparency are recognized and translated into accountability mechanisms" (Ciplet et al., 2018, p. 130), detailed information on bilateral initiatives as well as regional and national funds are often less readily available (Roberts et al., 2021; Watson & Schalatek, 2021; Weikmans & Roberts, 2019). Because of this, it is difficult to get accurate information on the accounting of fund distributions, leaving behind enduring challenges with accounting and transparency (Weikmans & Roberts, 2019; Weikmans et al., 2019).

These overarching concerns are captured in the 2014 SAMOA Pathway, the outcome document of the Third UN Conference on SIDS, as follows: (1) the international community, including its seven MAFs, has not mobilized sufficient capital to meet the costs of adaptation in SIDS; (2) many SIDS struggle to access MCF finance; and (3) the financial resources have not satisfied the Convention's obligations to enhance adaptive capacity:

> "… *financial resources available to date have not been adequate* to facilitate the implementation of climate change adaptation and mitigation projects, and … at times, *complex application procedures have prevented some [SIDS] from gaining access* to funds that are available internationally … [Further,] SIDS consider that the level of resources has been insufficient to ensure their capacity to respond effectively to multiple crises, and that without the necessary resources, *they have not fully succeeded in building capacity, strengthening national institutions according to national priorities*, *gaining access* and developing renewable energy and other environmentally sound technologies …" (SAMOA Pathway, 2014, emphases added).

Recent research shows that current flows, whether committed or disbursed, are only a fraction of the total costs of adaptation in SIDS. Regarding (1) (i.e., the international community has not mobilized sufficient capital to meet the costs of adaptation in SIDS), the sixth edition of the UN Environment Programme's Adaptation Gap Report, published in November 2021, had several key messages, including: "The finance needed to implement adaptation plans [in developing countries] is still far short of where it should be" and that "[n]ew estimated costs of adaptation are believed to be in the higher end of the cost brackets laid out in the Adaptation Gap Report 2016, when they were USD 140–300 billion per year by 2030 and USD 280–500 billion per year by 2050 for developing countries only" (UNEP, 2021, online). The figures we presented in Section 3.1 corroborate that adaptation finance flows to SIDS are significantly less than optimal.

Regarding (2) (i.e., many SIDS struggle to access MCF finance), Robinson and Dornan (2017, p. 1111) capture salient quotes from SIDS policy- and decision-makers about their experience with access—one interviewee explained, "[t]he GCF paperwork is higher than the sea-level rise in Tuvalu'', signaling the cumbersome nature of preparing project proposals. Another related concern for policy- and decision-makers is the type of finance to which SIDS have access. In the Caribbean, 38% of flows are concessional loans and 62%, grants (Atteridge et al., 2017); however, the situation in the AIMS region is starkly different—nearly 75% of the flows are in the form of concessional loans; grants account for the remaining 25% (Canales et al., 2017), raising questions about issues of fairness and justice for SIDS having to finance adaptation to climate impacts to which they made a negligible contribution. In the Pacific, 86% of finance is delivered as project-based support (Atteridge & Canales, 2017), raising questions about the sustainability of adaptation interventions. Furthermore, direct budget support has been rare (Atteridge & Canales, 2017), signaling the importance of works such as Rambarran (2018) that support cross-regional lesson-learning by, for example, showcasing the experience of Seychelles with successfully devising innovative financing mechanisms for supporting adaptation and conservation goals, all while reducing its public debt.

Another critical component of access is climate finance readiness. Ford and King (2015, p. 505) emphasize "readiness"—the degree

Table 2
Percentage of SIDS engaged in various adaptation actions.

Rank	Adaptation Type	% of Sampled SIDS Engaged	Rank	Adaptation Type	% of Sampled SIDS Engaged
1	Impact or vulnerability assessments	97.1%	12	Enforcement or follow-up of planning actions etc.	48.6%
2	Observation or climate change scenario modeling	88.6%	13	Knowledge management platform or network	45.7%
3	Public and oher stakeholder awareness and outreach	85.7%	14	Framework or policy instrument	42.9%
4	National committees or organizational changes	74.3%	15	Adaptation research	34.3%
5	Resource transfer	68.6%	16	Performance reviews/recommendations	25.7%
6	Work programs or strategic/action plans	60.0%	17	Draft policy or bill	22.9%
7	Infrastructure/technology/innovation	60.0%	18	Other implementation	22.9%
8	Other observation and assessment	57.1%	19	Other stakeholder engagement and knowledge management	14.3%
9	Participation in conferences, working groups, etc.	57.1%	20	Regulations, laws, statutes	8.6%
10	Surveillance or monitoring	54.3%	21	Financing mechanism	8.6%
11	Other planning	48.6%	22	Other monitoring and evaluation	5.7%

(Source: Robinson, 2020a, p. 9, reproduced with permission).

to which "human systems are prepared and ready to 'do adaptation'''—as an imperative for managing climate change. Conceptually, readiness includes the ability of systems to adapt *and* to receive and manage resources for adaptation. In 2012, The Nature Conservancy conducted a study in several developing countries that included three SIDS (Seychelles, St. Lucia, and Papua New Guinea). The study identified three key challenges of achieving climate finance readiness—(1) national strategies needing to chart a clear road map in order for climate change to be mainstreamed into development planning (TNC, 2012); (2) limited in-country coordination mechanisms creating dispersion and disorder in decision-making processes (TNC, 2012); and (3) investments in international climate cooperation, and related structures and arrangements, needing to be better aligned with ongoing institutional and policy developments (TNC, 2012). Following this work, several studies have largely suggested that these challenges remain, along with new and emerging ones. Samuwai and Hills (2018, p. 1), for example, concluded that there is a "massive readiness gap between countries in the Asian subregion and those in the Pacific sub-region". Among the SIDS sampled, Samoa exhibited the most progress with respect to policy and institutional readiness; and Tonga with respect to knowledge management and learning, and fiscal policy and management (Samuwai & Hills, 2018). Perhaps most recently, a study by Kiremu et al. (2022) found that the role of the private sector in climate finance readiness could be better recognized and supported, albeit in Kenya.

Regarding (3) (i.e., financial resources have not satisfied the Convention's obligations to enhance adaptive capacity), Robinson (2020a) finds considerable variability in adaptation progress and adaptive capacity improvements in SIDS. Compiling 1,497 adaptation actions from 35 SIDS across the three main geographic regions, Robinson (2020a) identifies five typologies of adaptation actions being pursued by SIDS (see Table 2 below): (1) observation and assessment (e.g., conducting vulnerability assessments); (2) planning (e.g., developing formal climate action plans or policies); (3) implementation (e.g., construction of infrastructure or measures of climate policy enforcement); (4) monitoring and evaluation (e.g., performance review sessions); and (5) stakeholder engagement and knowledge management (e.g., climate change education forums for public audiences). Robinson (2020a) identifies 'more advanced adaptors', or countries performing many adaptation actions. Cook Islands and Kiribati in the Pacific head the 'more advanced adaptors' list. Robinson (2020a) also identifies 'less advanced adaptors', or countries that perform few adaptation actions. Guinea-Bissau in the AIMS region and the Marshall Islands in the Pacific head the 'less advanced adaptors' list. Dominica in the Caribbean falls in between the two groups, straddling the line between 'more advanced adaptors' and 'less advanced adaptors'. Of special note are the discrepancies between the kinds of adaptation strategies that the more and less advanced adaptors prioritize. On average, more advanced adaptors have concentrated on 'planning', 'implementation', and 'monitoring/evaluation'; less advanced adaptors have pursued 'observation and assessment' and 'stakeholder engagement/knowledge management' (Robinson, 2020a), suggesting that most SIDS are engaged in what Berrang-Ford et al. (2014) call 'groundwork' activities that precede 'actual' adaptation. There are also regional variations that are worth noting: AIMS SIDS are strongest in 'stakeholder engagement and knowledge management'; Caribbean SIDS in 'observation and assessment of climate variables'; and Pacific SIDS in 'planning', 'implementation', and 'monitoring/evaluation' (Robinson, 2020a). Much of this can be credited to the strength of regional programming or lack thereof as it is well established in the literature that SIDS pool resources at the regional level to minimize the challenges associated with capacity constraints (see Gilfillan et al., 2020; Kelman, 2016; Robinson & Gilfillan, 2017; Robinson et al., 2022).

4.1. Funding levels are a byproduct of divergences in operationalizing equity-oriented norms

Another challenge for mobilizing adaptation finance through MAFs, in particular, has been operationalizing an *"appropriate burden sharing"* mechanism among developed country Parties, as stipulated by the UNFCCC and subsequent agreements. Equity-oriented norms such historical responsibility and common but differentiated responsibilities and respective capabilities, are promoted to guide the fundraising architecture of MCFs (Pickering et al., 2017)—the UN Secretary-General recently called on all donors to increase the share of adaptation and resilience finance to at least 50% of climate finance support (United Nations, 2021b). Yet, (1) there is no objective method for measuring historical responsibility, and (2) it is difficult to disentangle historical and current contributions. This is the case mainly because historical emissions often have long residence times and extend long into the future (Weaver et al., 2007), and a number of modern-day emitters such as China have not traditionally assumed the responsibility to lead climate action globally (Carlson et al., 2021). As a result, different financial entities use vastly different burden sharing schemes, which, again, are complicated by the fragmentation problem. A study by Cui and Huang (2018, p. 173), for example, revealed "that the ongoing international financing mechanisms vary in their burden sharing results and the shares of existing donors are driven by highly complex reasons"—such as the ability of the GEF to *actually* deliver capacity building finance (also see Stender et al., 2020), and official development assistance being channeled through the OECD, which faces mounting challenges to attracting private sector investments in adaptation (also see Moser et al., 2019). These and similar studies also show that burden sharing is sensitive to the participation of and contributions from developed countries, demonstrating the importance of cooperation to the success of financial mechanisms.

In addition to complications with mobilizing funds, the multiplicity of MCF institutions, each with its respective interests, mandates, fiduciary standards, and decision-making arrangements, exacerbates the issue of SIDS' access to funding, as it imposes administrative burdens on those countries that have governance systems that are already burdened (Gomez-Echeverri, 2013; Hurley, 2015; Pickering et al., 2017). In Caribbean SIDS, for example, Robinson (2018c) argued that competing development priorities such as poverty reduction, healthcare and crime fighting reduced governments' fiscal space to tackle adaptation. Additionally, SIDS' access-related difficulties have been increasingly recognized in international fora (e.g., the 2014 SAMOA Pathway) and in the academic and policy literatures. Tortora and Soares (2016) identified challenges of Caribbean SIDS' access to funding, many of which were institutional (e.g., policy constraints); other challenges included donors' myopic finance allocation priorities (e.g., providing resilience funding for post-disaster recovery, but no money for long-term needs), fragmented delivery of finance, and the voluminous application paperwork required by MCFs.

As an example, the Inter-American Development Bank has aimed to dedicate at least 35% of its annual lending approval volume to the small and vulnerable countries in the Latin America and Caribbean region. However, only nine SIDS—Bahamas, Barbados, Belize, Dominican Republic, Guyana, Haiti, Jamaica, Suriname, and Trinidad and Tobago—are borrowing member countries (IDB, 2021a; 2022a). Through a unique charter provision, Bank resources can also be channeled through the Caribbean Development Bank to non-member SIDS that are members of the Organisation of Eastern Caribbean States (IDB, 2017). In this way, and as an example, funds have been channeled through a US$50 million Global Credit Loan to the Caribbean Development Bank to support enhancing the disaster resilience of infrastructure and foster disaster-resilient growth for micro, small, and medium-sized enterprises in Antigua and Barbuda, Dominica, Grenada, St. Kitts and Nevis, St. Lucia, and St. Vincent and the Grenadines (IDB, 2021b). While making funds available to these non–Inter-American Development Bank member countries, the arrangement also creates an administrative and financial burden as these countries would also need membership in the Organisation of Eastern Caribbean States and the Caribbean Development Bank in order to access the funds. In order to become a regional member of the Inter-American Development Bank, a country needs prior membership in the Organization of American States, and a subscription of shares of the Ordinary Capital along with a contribution to the Fund for Special Operations (IDB, 2022b). As a result, many SIDS are unable to accrue ample financing from MCFs, which explains why SIDS are increasingly reliant on bilateral adaptation finance from bilateral official development assistance through OECD Member States (Petzold & Magnan, 2019; Robinson & Dornan, 2017; Thomas et al., 2019). This is complemented by a push for more South-South and triangular cooperation where adaptation projects and programs in SIDS are supported technically and financially by other developing countries, including other SIDS (e.g. see Rhiney, 2018; Yairen, 2022).

4.2. Funding access is a byproduct of divergences in allocation criteria

Another ethical dilemma related to climate finance is identifying which countries are “particularly vulnerable” and thus should be prioritized for receiving funds. While SIDS are internationally recognized as particularly vulnerable, there is no consensus on the list of countries that should be classified as “highly vulnerable” (Weikmans et al., 2017, p. 7). The GCF, for example, earmarks half of its adaptation finance for the most vulnerable countries—African countries, LDCs and SIDS. Yet, because of a lack of clarity around how this is defined and possible within-group variations, Brechin and Espinoza (2017) and others have called for the allocation strategy not to be arbitrary. This has led other multilaterals to revisit their metrics. The Caribbean Development Bank, for example, recently revised its Vulnerability Index and widened the scope of previous estimations to include various climate change components along with social vulnerability. Among the main findings were that borrowing member countries, on average, can be considered middle-to-high vulnerability countries, and that vulnerabilities primarily revolve around dependence on a few major export products and trading partners, high levels of energy imports and related products, social challenges such as crime, and exposure to natural hazards and climate change (Ram et al., 2019). Although resource-rich Caribbean SIDS such as Trinidad and Tobago, and Suriname had lower vulnerability scores, the report noted that these economies nevertheless “have unique challenges related to sharp boom and bust cycles and higher levels of inequality” (Ram et al., 2019, p. 9). It would, therefore, be reasonable to expect that the Caribbean Development Bank will prioritize those countries with the highest vulnerability, such as Haiti, Jamaica and St. Lucia, but this remains unclear.

The susceptibility of vulnerability to differing interpretations likely explains, at least in part, the geographic trends in the adaptation progress of SIDS, with most MAF finance going to the Pacific (Robinson & Dornan, 2017). Some scholars argue that the ways in which financial entities operationalize ‘vulnerability’ in funding decisions are often premised on misguided conceptions of vulnerability (Bourne et al., 2015; Horstmann, 2011). For example, Persson and Remling (2014) explored alternative funding allocation rationales based on equity and efficiency concerns, and tested them against the criteria and priorities of the Adaptation Fund. Their findings suggest that equity concerns are a primary motivation for the Fund’s allocation decisions, but only when comparing the vulnerability of States (i.e., at the national or federal level) and not subnational communities. This conclusion, though later challenged by the Adaptation Fund Board, has implications for access to funds. To illustrate this point, consider Bourne et al. (2015), who argue against the use of economic vulnerability indices that are based on per capita income alone. Analyses of official development assistance flows to Caribbean SIDS demonstrate substantial decreases over time due to donors’ finance preference shifting towards lower-income countries, rendering most Caribbean SIDS unable to meet the eligibility criteria for various funding from sources of development finance (for adaptation) (Bourne et al., 2015). Yet, Caribbean SIDS allocate less than 4% of fiscal expenditures to capital investment due to large amounts of debt, which is below investment requirements for, *inter alia*, adaptation and the expansion of economic and social infrastructure (Bourne et al., 2015), which are required to progress countries towards sustainable development pathways.

Another allocation criterion used by MAFs is project size. Unlike the GEF, the GCF does not fund small pilot projects, and so, funding proposals fall into one of four project sizes: <US$10 million (micro), US$10–50 million (small), US$50–250 (medium), or > US$250 million (large). To date, only 11% of approved GCF projects have been micro projects (GCF, 2022). On the one hand, only two micro projects have been approved to support adaptation in SIDS—US$2.3 million in GCF funding (with no co-financing) to enhance climate resilience in the water sector in Bahrain, and US$8.6 million in GCF funding (with a total project value of US$9.4 million) to support climate resilient food security for farming households in the Federated States of Micronesia (GCF, 2022). Several small adaptation projects have been approved for SIDS, including Antigua and Barbuda, Cuba, Marshall Islands, Timor-Leste, and Vanuatu. On the other hand, some Caribbean SIDS, such as Dominica, St. Kitts and Nevis, St. Lucia and St. Vincent and Grenadines, have only been able to access GCF funds through a multi-country initiative—the Sustainable Energy Facility for the Eastern Caribbean, a mitigation project geared towards addressing the financial, technical and institutional barriers to geothermal energy development in the sub-region (GCF, 2022).

5. Reorienting multilateral adaptation finance towards resilience

Supporting previous calls from Bowman and Minas (2019) and others, a reorientation of multilateral adaptation finance towards resilience could help to overcome some of its pitfalls and to recenter flows on the foundational principles of climate finance—i.e., on "the basis of equity and in accordance with their common but differentiated responsibilities and respective capabilities" (UNFCCC, 1992, online). The concept of resilience, like the other concepts guiding the climate change regime, is broad, but is generally concerned with a system's, community's, organization's, economy's, or other entity's response to perturbations. Specifically, it concerns the abilities to anticipate and plan for climate change (in this case), sustain and quickly recover from its impacts, and adapt to future changes (Folke, 2006; Miller et al., 2010; Smit & Wandel, 2006). Resilience also has within its scope the roles of institutions, leadership, social capital, knowledge and resource sharing, reflexivity, and adaptive governance, which are critical institutional qualities for enhancing it (Folke, 2006; Holling, 1978; Miller et al., 2010; Sellberg et al., 2018).

The first source of resilience's utility lies within these latter qualities, conceived here as qualities contributing to adaptive capacity, which we understand as being a component of resilience. Improving marginalized and vulnerable communities' access to international adaptation financing must be seen as necessitating facilitation of their adaptive capacity, as adaptive capacity undoubtedly has implications for the ability of countries to access funds. Hence, the climate change regime could employ resilience as a guiding principle and/or as one of MAFs' funding priorities. The objectives of newer MAFs, like the Global Climate Change Alliance, which aims to strengthen dialogue—i.e., facilitate the resilience quality of knowledge sharing—are already indicative of the growing recognition of resilience as an imperative. Though SIDS policy- and decision-makers find the term ambiguous and, thereby, difficult to operationalize (see Saxena et al., 2018), this push could be especially relevant to SIDS, given their limited institutional capacities to plan for, respond to and sustain, and adapt to future climate change.

The comprehensiveness of resilience also offers potential opportunities for the integration of agendas. For example, resilience, as described above, captures both adaptation and disaster risk reduction within its scope. The potential mainstreaming of agendas, as encouraged by resilience, is already being pursued—take the 2007 Bali Action Plan and the 2010 Cancun Adaptation Framework, both of which were agreed within the context of the UNFCCC and call for more recognition of disaster risk reduction as part of the climate adaptation agenda, or for more on-the-ground initiatives. Vanuatu in the Pacific region, for example, has developed an integrated national draft policy on climate change and disaster risk reduction (Robinson, 2019). Additionally, a new Pacific-wide integrated strategy is currently being drafted that will integrate the currently isolated Disaster Management Framework for Action (2006–2015) and the Pacific Islands Framework for Action on Climate Change (2006–2015) (Robinson, 2019). Resilience may even serve to coordinate the agendas of mitigation *and* adaptation, e.g., via the restoration of blue carbon ecosystems (e.g. see Robinson et al., 2021). Many SIDS are reliant on blue carbon ecosystems for both revenue (financial resilience) and storm surge protection (coastal resilience) (Mycoo, 2018). Channeling funds to blue carbon restoration would ensure continued resilience *and* enhanced mitigation given the sink functionality these ecosystems provide. Additionally, there is already evidence of some progress in Seychelles in the AIMS region (see Chevallier et al., 2019) and in Vanuatu (see Sahin et al., 2021). However it may be achieved, the harmonization and/or reorientation of the funding priorities of the climate change regime constitute imperative first steps to addressing the pitfalls of MAFs and other climate finance entities, as demonstrated by the SIDS experience.

6. Conclusion

This paper is a narrative review of several key issue areas of MCFs and the climate finance landscape more broadly. Using the experience of SIDS, we reviewed the nature and pitfalls of multilateral adaptation finance. Regarding the nature of finance, we found that (1) sources and flows of adaptation finance are highly skewed across countries, and (2) adaptation finance is being channeled towards 'general environmental protection'. While identifying where the overarching problems lie is a challenging task, especially in view of the vastness and interrelatedness of the issues plaguing MCFs, we deduced that (3) countries receiving the most finance are not the most advanced adaptors, but that (4) funding levels are a byproduct of divergences in the operationalization of equity-oriented norms, and (5) funding access is a byproduct of divergences in allocation criteria. The main conclusion from our effort is that, because MCFs variedly operationalize the principles underlying climate finance, inadequacies and barriers for SIDS and other particularly vulnerable countries are created.

Going forward, in order to better meet country needs, MCFs would need to be more flexible, less risk averse, and more transparent in the ways in which they report the results achieved and the overall impact of international public finance (Nakhooda et al., 2014). There is also a need for a robust and mandatory metadata system that accurately tracks flows of finance into SIDS and other particularly vulnerable countries for adaptation—the system would aim to standardise what counts as 'adaptation' along with definitions of 'adaptation finance' etc., and facilitate independent, third party analyses of financial flows (e.g. see Donner et al., 2016). Closing the climate finance readiness gaps across SIDS will also be important. However, as work by Samuwai and Hills (2018) shows, there are massive gaps across countries, suggesting that improving readiness alone is insufficient for unlocking adaptation finance and its transformational potential. As a result, attention must also be paid to complementary areas such as governance and the diversification of flows.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work did not receive any specific grants.

References

Adaptation Fund, 2014. NIE Accreditation Toolkit. Retrieved from: https://www.adaptation-fund.org/wp-content/uploads/2015/07/Accreditation-Toolkit-English-14.pdf. Accessed March 7, 2021.

Adeoti, T., Boroto, D., Haggis, R., Thacker, S., Felix, F., Edwin, G., & Wells, H., 2021. Saint Lucia: National Infrastructure Financing Strategy. United Nations Office for Project Services, Copenhagen, Denmark. Retrieveed from: https://content.unops.org/publications/Saint-Lucia-National-Infrastructure-Financing-Strategy.pdf. Accessed March 12, 2022.

African Development Bank (AfDB), 2014. Joint Report on MDB Climate Finance. Retrieved from: http://www.eib.org/attachments/documents/joint_report_on_mdb_climate_finance_2013.pdf. Accessed March 7, 2021.

Ameli, N., Drummond, P., Bisaro, A., Grubb, M., Chenet, H., 2020. Climate finance and disclosure for institutional investors: why transparency is not enough. Clim. Change 160 (4), 565–589. https://doi.org/10.1007/s10584-019-02542-2.

Atteridge, A., & Canales, N., 2017. Climate finance in the Pacific: An overview of flows to the region's Small Island Developing States. Stockholm Environment Institute: Working Paper 2017-04. Retrieved from: https://www.greengrowthknowledge.org/sites/default/files/downloads/resource/SEI-WP-2017-04-Pacific-climate-finance-flows.pdf. Accessed September 7, 2021.

Atteridge, A., Canales, N., & Savvidou, G., 2017. Climate finance in the Caribbean region's Small Island Developing States. Stockholm Environment Institute: Working Paper 2017-2018. Retrieved from: https://www.sei.org/publications/caribbean-climate-finance/. Accessed September 7, 2021.

Ayers, J.M., Abeysinghe, A.C., 2013. International aid and adaptation to climate change. Handbook Global Clim. Environ. Policy 486–506.

Berrang-Ford, L., Ford, J.D., Lesnikowski, A., Poutiainen, C., Barrera, M., Heymann, S.J., 2014. What drives national adaptation? A global assessment. Clim. Change 124 (1–2), 441–450. https://doi.org/10.1007/s10584-014-1078-3.

Bertilsson, J., Thörn, H., 2021. Discourses on transformational change and paradigm shift in the Green Climate Fund: the divide over financialization and country ownership. Environ. Politics 30 (3), 423–441. https://doi.org/10.1080/09644016.2020.1775446.

Betzold, C., Weiler, F., 2017. Allocation of aid for adaptation to climate change: Do vulnerable countries receive more support? Int. Environ. Agreements: Politics, Law Econ. 17 (1), 17–36. https://doi.org/10.1007/s10784-016-9343-8.

Bourne, C., Alexander, M., Conrad, D., & Jhinkoo, J., 2015. Financing for development challenges in Caribbean SIDS: A case for review of eligibility criteria for access to concessional financing. Retrieved from http://www.undp.org/content/dam/rblac/docs/Research%20and%20Publications/Poverty%20Reduction/UNDP_RBLAC_Financing_for_Development_ReportCaribbean.pdf. Accessed June 21, 2021.

Bowman, M., Minas, S., 2019. Resilience through interlinkage: the green climate fund and climate finance governance. Climate Policy 19 (3), 342–353. https://doi.org/10.1080/14693062.2018.1513358.

Bracking, S., Leffel, B., 2021. Climate finance governance: fit for purpose? WIREs Clim. Change 12 (4), e709.

Brechin, S.R., Espinoza, M.I., 2017. A case for further refinement of the green climate fund's 50:50 ratio climate change mitigation and adaptation allocation framework: toward a more targeted approach. Clim. Change 142 (3–4), 311–320. https://doi.org/10.1007/s10584-017-1938-8.

Brown, J., Kaur, N., 2009. Financing adaptation: matching form with function. ODI Background Note. Overseas Development Institute, London.

Brunner, S., Enting, K., 2014. Climate finance: a transaction cost perspective on the structure of state-to-state transfers. Global Environ. Change 27, 138–143. https://doi.org/10.1016/j.gloenvcha.2014.05.005.

Buchner, B., Clark, A., Falconer, A., Macquarie, R., Meattle, C., Tolentino, R., & Wetherbee, C., 2019. The Global Landscape of Climate Finance 2019. Retrieved from: https://bit.ly/3m01LwB. Accessed June 21, 2021.

Bulman, D., Kolkma, W., Kraay, A., 2017. Good countries or good projects? Comparing macro and micro correlates of World Bank and Asian Development Bank project performance. Rev. Int. Org. 12 (3), 335–363. https://doi.org/10.1007/s11558-016-9256-x.

Cabezon, E., Hunter, L., Tumbarello, P., Washimi, K., Wu, Y., 2019. Enhancing macroeconomic resilience to natural disasters and climate change in the small states of the Pacific. Asian-Pacif. Econ. Literature 33 (1), 113–130. https://doi.org/10.1111/apel.12255.

Canales, N., et al., 2017. Climate finance for the Indian Ocean and African Small Island Developing States. Retrieved from https://bit.ly/2lLsjqn. Accessed March 7, 2021.

Carlson, D.A., Robinson, S.-A., Blair, C., McDonough, M., 2021. China's climate ambition: revisiting its First Nationally Determined Contribution and centering a just transition to clean energy. Energy Policy 155, 112350. https://doi.org/10.1016/j.enpol.2021.112350.

Chen, L., 2018. The challenges and opportunities for the green climate fund. Chinese J. Urban Environ. Stud. 06 (01), 1875001. https://doi.org/10.1142/s2345748118750015.

Chevallier, R., Bijoux, J., Macamo, C., & Sauka, S., 2019. Marine and Coastal Ecosystem-based Adaptation for Enhanced Resilience in Southern Africa: Synthesis Report. Retrieved from https://bit.ly/3nXbnLm. Accessed November 27, 2021.

Ciplet, D., Adams, K.M., Weikmans, R., Roberts, J.T., 2018. The transformative capability of transparency in global environmental governance. Global Environ. Politics 18 (3), 130–150. https://doi.org/10.1162/glep_a_00472.

Commonwealth Secretariat, 2021. Belize explores debt-for-climate swaps with Commonwealth support. Retrieved from: https://thecommonwealth.org/news/belize-explores-debt-climate-swaps-commonwealth-support. Accessed March 12, 2022.

Cui, L., Huang, Y., 2018. Exploring the schemes for green climate fund financing: international lessons. World Dev. 101, 173–187. https://doi.org/10.1016/j.worlddev.2017.08.009.

Cui, L., Sun, Y., Song, M., Zhu, L., 2020. Co-financing in the green climate fund: lessons from the global environment facility. Climate Policy 20 (1), 95–108. https://doi.org/10.1080/14693062.2019.1690968.

Dessai, S., 2001. Prof–le - why did the Hague climate conference fail? Environ. Politics 10 (3), 139–144. https://doi.org/10.1080/714000552.

Donner, S.D., Kandlikar, M., Webber, S., 2016. Measuring and tracking the flow of climate change adaptation aid to the developing world. Environ. Res. Lett. 11 (5), 054006.

European Bank for Reconstruction and Development, African Development Bank, Asian Development Bank, European Investment Bank, Inter-American Development Bank Group, Islamic Development Bank, & World Bank Group. (2017). Joint Report on Multilateral Development Banks' Climate Finance. Retrieved from https://bit.ly/2JGgXv6. Accessed September 23, 2019.

Fonta, W.M., Ayuk, E.T., van Huysen, T., 2018. Africa and the Green Climate Fund: current challenges and future opportunities. Clim. Policy 18 (9), 1210–1225. https://doi.org/10.1080/14693062.2018.1459447.

Ford, J.D., King, D., 2015. A framework for examining adaptation readiness. Mitigat. Adaptation Strategies Global Change 20 (4), 505–526. https://doi.org/10.1007/s11027-013-9505-8.

Garschagen, M., Doshi, D., 2022. Does funds-based adaptation finance reach the most vulnerable countries? Global Environ. Change 73, 102450. https://doi.org/10.1016/j.gloenvcha.2021.102450.

Green Climate Fund (GCF), 2020. Independent Evaluation of the Relevance and Effectiveness of the Green Climate Fund's Investments in Small Island Developing States. Retrieved from https://ieu.greenclimate.fund/sites/default/files/evaluation/201123-sids-final-report-ex-sum-top.pdf. Accessed June 21, 2021.

Green Climate Fund (GCF), 2021. About GCF. Retrieved from https://www.greenclimate.fund/about/timeline#:~:text=GCF%20was%20established%20under%20the,UNFCCC%20and%20the%20Paris%20Agreement. Accessed November 20, 2021.

Green Climate Fund (GCF), 2022. Project portfolio. Retrieved from: https://www.greenclimate.fund/projects. Accessed March 12, 2022.

Geddes, A., Schmidt, T., Steffen, B., 2018. The multiple role of state investment banks in low-carbon energy finance: an analysis of Australia, the UK and Germany. Energy Policy 115, 158–170. https://doi.org/10.1016/j.enpol.2018.01.009.

Giglio, S., Kelly, B., Stroebel, J., 2021. Climate finance. Annu. Rev. Financial Econ. 13 (1), 15–36. https://doi.org/10.1146/annurev-financial-102620-103311.

Gilfillan, D., Robinson, S.-A., Barrowman, H., 2020. Action research to enhance inter-organisational coordination of climate change adaptation in the pacific. Challenges 11 (8), 1–23. https://doi.org/10.3390/challe11010008.

Gomez-Echeverri, L., 2013. The changing geopolitics of climate change finance. Clim. Policy 13 (5), 632–648. https://doi.org/10.1080/14693062.2013.822690.

Hall, N., 2017. What is adaptation to climate change? Epistemic ambiguity in the climate finance system. Int. Environ. Agreements: Politics, Law Econ. 17 (1), 37–53. https://doi.org/10.1007/s10784-016-9345-6.

Horstmann, B., 2011. Operationalizing the Adaptation Fund: challenges in allocating funds to the vulnerable. Clim. Policy 11 (4), 1086–1096. https://doi.org/10.1080/14693062.2011.579392.

Hurley, G., 2015. Financing for development and small island developing states: A snapshot and ways forward. UNDP and UN-OHRLLS Discussion Paper. New York: United Nations Development Programme and The United Nations Office of the High Representative for Least Developed Countries, Landlocked Developing Countries and Small Island Developing States. Retrieved from: https://sustainabledevelopment.un.org/index.php?page=view&type=400&nr=2181&menu=35. Accessed November 20, 2021.

Inter-American Development Bank (IDB), 2017. CDB, IDB sign agreement to strengthen partnership. Retrieved from: https://www.iadb.org/en/news/cdb-idb-sign-agreement-strengthen-partnership. Accessed March 12, 2022.

Inter-American Development Bank (IDB), 2021. Borrowing Member Countries. Retrieved from: https://www.iadb.org/en/about-us/borrowing-member-countries. Accessed March 12, 2022.

Inter-American Development Bank (IDB), 2021. Caribbean Development Bank to Enhance the Climate Resilience of Caribbean with IDB support. Retrieved from: https://www.iadb.org/en/news/caribbean-development-bank-enhance-climate-resilience-caribbean-idb-support. Accessed March 12, 2022.

Inter-American Development Bank (IDB), 2022. How We Are Organized. Retrieved from: https://www.iadb.org/en/about-us/how-are-we-organized. Accessed March 12, 2022.

Inter-American Development Bank (IDB), 2022. Capital and Funds Under Administration. Retrieved from: https://www.iadb.org/en/idb-finance/capital-and-funds-under-administration. Accessed March 12, 2022.

International Institute for Environment and Development (IIED), 2021. Least developed countries unable to borrow over 80 per cent of money needed to adapt to climate change. Retrieved from: https://www.iied.org/least-developed-countries-unable-borrow-over-80-cent-money-needed-adapt-climate-change. Accessed March 12, 2022.

Kasdan, M., Kuhl, L., Kurukulasuriya, P., 2021. The evolution of transformational change in multilateral funds dedicated to financing adaptation to climate change. Clim. Dev. 13 (5), 427–442. https://doi.org/10.1080/17565529.2020.1790333.

Kelman, I., 2016. Governance of climate change adaptation on Small Island Developing States (SIDS). In: Knieling, J. (Ed.), Climate Adaptation Governance in Cities and Regions: Theoretical Fundamentals and Practical Evidence. Wiley Blackwell, Hoboken, NJ, pp. 355–369.

Kiremu, M., Scrimgeour, F., Mutegi, J., Mumo, R., 2022. Climate finance readiness: a review of institutional frameworks and policies in Kenya. Sustain. Environ. 8 (1), 2022569. https://doi.org/10.1080/27658511.2021.2022569.

Lattanzio, R. K., 2014. International climate change financing: the green climate fund (GCF): Congressional Research Service. Retrieved from: https://sgp.fas.org/crs/misc/R41889.pdf. Accessed November 20, 2021.

Liverman, D., Billett, S., 2010. Copenhagen and the Governance of Adaptation. Environ.: Sci. Policy Sustain. Dev. 52 (3), 28–36. https://doi.org/10.1080/00139151003761579.

Moser, S.C., Ekstrom, J.A., Kim, J., Heitsch, S., 2019. Adaptation finance archetypes: local governments' persistent challenges of funding adaptation to climate change and ways to overcome them. Ecol. Society 24 (2). https://doi.org/10.5751/ES-10980-240228.

Munira, S., Bashar, R., Easher, T.H., Khan, M.R., 2021. Climate Finance in the UNFCCC Negotiations: Bridging Gaps with Lessons Learnt. In: Jakariya, M., Islam, M.N. (Eds.), Climate Change in Bangladesh: A Cross-Disciplinary Framework. Springer International Publishing, Cham, pp. 1–24.

Mycoo, M.A., 2018. Beyond 1.5°C: vulnerabilities and adaptation strategies for Caribbean Small Island Developing States. Reg. Environ. Change 18 (8), 2341–2353. https://doi.org/10.1007/s10113-017-1248-8.

Mycoo, M., et al., 2022. Small islands. In: Pörtner, H.O. (Ed.), Climate Change 2022: Impacts, Adaptation, and Vulnerability. Part B: Regional Aspects. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, pp. 2660–2766.

Nakhooda, S., Norman, M., Barnard, S., Watson, C., Greenhill, R., Caravani, A., Banton, G., 2014. Climate Finance: is it making a difference? A review of the effectiveness of the Multilateral Climate Funds. Overseas Development Institute, London.

Nelson, D. & Pierpont, B., 2013. The challenge of institutional investment in renewable energy. Climate Policy Initiative report. Retrieved from: http://climatepolicyinitiative.org/wp-content/uploads/2013/03/The-Challenge-of-Institutional-Investment-in-Renewable-Energy.pdf. Accessed March 12, 2022.

Nelson, R. M., 2012 Multilateral development banks: Overview and issues for Congress. Retrieved from: https://sgp.fas.org/crs/row/R41170.pdf. Accessed October 20, 2021.

Nunn, P.D., Aalbersberg, W., Lata, S., Gwilliam, M., 2013. Beyond the core: community governance for climate-change adaptation in peripheral parts of Pacific Island Countries. Reg. Environ. Change 14 (1), 221–235. https://doi.org/10.1007/s10113-013-0486-7.

Nunn, P.D., Klöck, C., Duvat, V., 2021. Seawalls as maladaptations along island coasts. Ocean Coast. Manage. 205, 105554. https://doi.org/10.1016/j.ocecoaman.2021.105554.

Nurse, L.A., McLean, R.F., Agard, J., Briguglio, L.P., Duvat-Magnan, V., Pelesikoti, N., Tompkins, E., Webb, A., 2014. Small islands. In: Barros, V.R., Field, C.B., Dokken, D.J., Mastrandrea, M.D., Mach, K.J., Bilir, T.E., Chatterjee, M., Ebi, K.L., Estrada, Y.O., Genova, R.C., Girma, B., Kissel, E.S., Levy, A.N., MacCracken, S., Mastrandrea, P.R., White, L.L. (Eds.), Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part B: Regional Aspects. Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press, Cambridge, pp. 1613–1654.

Omari-Motsumi, K., Barnett, M., & Schalatek, L., 2019. Broken Connections and Systemic Barriers: Overcoming the Challenge of the "Missing Middle" in Adaptation Finance. Global Commission on Adaptation Background Paper. Retrieved from: https://gca.org/reports/broken-connections-and-systemic-barriers-overcoming-the-challenge-of-the-missing-middle-in-adaptation-finance/. Accessed March 15, 2021.

Persson, A., Remling, E., 2014. Equity and efficiency in adaptation finance: initial experiences of the Adaptation Fund. Clim. Policy 14 (4), 488–506. https://doi.org/10.1080/14693062.2013.879514.

Petzold, J., Magnan, A.K., 2019. Climate change: thinking small islands beyond Small Island Developing States (SIDS). Clim. Change 152 (1), 145–165. https://doi.org/10.1007/s10584-018-2363-3.

Pickering, J., Betzold, C., Skovgaard, J., 2017. Managing fragmentation and complexity in the emerging system of international climate finance. Int. Environ. Agreements 17, 1–16. https://doi.org/10.1007/s10784-016-9349-2.

Piggott-McKellar, A.E., Nunn, P.D., McNamara, K.E., Sekinini, S.T., 2020. Dam(n) Seawalls: A Case of Climate Change Maladaptation in Fiji. In: Filho, W.L. (Ed.), Managing Climate Change Adaptation in the Pacific Region. Springer International Publishing, Cham, pp. 69–84.

Rajamani, L., 2010. The making and unmaking of the Copenhagen Accord. Int. Comp. Law Q. 59 (3), 824–843. https://doi.org/10.1017/S0020589310000400.

Ram, J., Cotton, J. J., Frederick, R., & Elliott, W., 2019. Measuring Vulnerability: A Multidimensional Vulnerability Index for the Caribbean. Retrieved from: https://www.caribank.org/publications-and-resources/resource-library/working-papers/measuring-vulnerability-multidimensional-vulnerability-index-caribbean. Accessed March 13, 2022.

Rambarran, J., 2018. Debt for climate swaps: Lessons for Caribbean SIDS from the Seychelles' experience. Social and Economic Studies, 67(2-3), 261-291. Retrieved from https://bit.ly/30KoMIX. Accessed May 20, 2020.

Rhiney, K., 2018. Climate change and the future of agriculture in the Caribbean: Prospects for South-South cooperation. In: Fiddian-Qasmiyeh, E., Daley, P. (Eds.), Routledge Handbook of South-South Relations. Routledge, London.

Roberts, J.T., Weikmans, R., Robinson, S.-A., Ciplet, D., Khan, M., Falzon, D., 2021. Rebooting a failed promise of climate finance. Nature Clim. Change 11 (3), 180–182. https://doi.org/10.1038/s41558-021-00990-2.

Robinson, S.-A., 2017. Climate change adaptation trends in small island developing states. Mitigat. Adaptation Strategies Global Change 22 (4), 669–691. https://doi.org/10.1007/s11027-015-9693-5.

Robinson, S.-A., 2018a. Climate change adaptation in small island developing states: Insights and lessons from a meta-paradigmatic study. Environ. Sci. Policy 85, 172–181. https://doi.org/10.1016/j.envsci.2018.03.030.

Robinson, S.-A., 2018b. Climate change adaptation limits in small island developing states. In: Filho, W.L., Nalau, J. (Eds.), Limits to Climate Change Adaptation. Springer International Publishing, Cham, pp. 263–281.

Robinson, S.-A., 2018c. Adapting to climate change at the national level in Caribbean small island developing states. Island Stud. J. 13, 79–100. https://doi.org/10.24043/isj.59.

Robinson, S.-A., 2019. Mainstreaming climate change adaptation in small island developing states. Clim. Dev. 11 (1), 47–59. https://doi.org/10.1080/17565529.2017.1410086.

Robinson, S.-A., 2020a. A richness index for baselining climate change adaptations in small island developing states. Environ. Sustain. Indicators 8 (100065). https://doi.org/10.1016/j.indic.2020.100065.

Robinson, S.-A., 2020b. Climate change adaptation in SIDS: A systematic review of the literature pre and post the IPCC Fifth Assessment Report. Wiley Interdiscip. Rev.: Climate Change 11 (4), 1–21. https://doi.org/10.1002/wcc.653.

Robinson, S.-A., Dornan, M., 2017. International financing for climate change adaptation in small island developing states. Reg. Environ. Change 17 (4), 1103–1115. https://doi.org/10.1007/s10113-016-1085-1.

Robinson, S.-A., Gilfillan, D., 2017. Regional organisations and climate change adaptation in small island developing states. Reg. Environ. Change 17 (4), 989–1004. https://doi.org/10.1007/s10113-016-0991-6.

Robinson, S.-A., Carlson, D.A., Messer, A., Maunus, L., Bouton, E., Roberts, J.T., 2021. Climate compatible development in practice. Dev. Practice 1–11. https://doi.org/10.1080/09614524.2021.1907534.

Robinson, S.-A., Bouton, E., Dolan, M., Meakem, A., Messer, A., Lefond, I., Roberts, J.T., 2022. A new framework for rapidly assessing national adaptation policies: an application to small island developing states in the Atlantic and Indian Oceans. Reg. Environ. Change 22 (2). https://doi.org/10.1007/s10113-021-01855-2.

Sahin, O., Hadwen, W.L., Buckwell, A., Fleming, C., Ware, D., Smart, J.C.R., Mackey, B., 2021. Assessing how ecosystem-based adaptations to climate change influence community wellbeing: a Vanuatu case study. Reg. Environ. Change 21 (4), 90. https://doi.org/10.1007/s10113-021-01809-8.

Samuwai, J., Hills, J.M., 2018. Assessing climate finance readiness in the Asia-Pacific region. Sustainability 10 (1192), 1–18. https://doi.org/10.3390/su10041192.

Saxena, A., Qui, K., Robinson, S.-A., 2018. Knowledge, attitudes and practices of climate adaptation actors towards resilience and transformation in a 1.5°C world. Environ. Sci. Policy 80, 152–159. https://doi.org/10.1016/j.envsci.2017.11.001.

Schulz, K., Feist, M., 2021. Leveraging blockchain technology for innovative climate finance under the Green Climate Fund. Earth Syst. Governance 7, 100084. https://doi.org/10.1016/j.esg.2020.100084.

Stender, F., Moslener, U., Pauw, W.P., 2020. More than money: does climate finance support capacity building? Appl. Econ. Lett. 27 (15), 1247–1251. https://doi.org/10.1080/13504851.2019.1676384.

Thomas, A., Shooya, O., Rokitzki, M., Bertrand, M., Lissner, T., 2019. Climate change adaptation planning in practice: insights from the Caribbean. Reg. Environ. Change 19 (7), 2013–2025. https://doi.org/10.1007/s10113-019-01540-5.

The Nature Conservancy (TNC), 2012. Climate Finance Readiness: Lessons Learned in Developing Countries. Retrieved from: https://www.nature.org/media/climatechange/climate-finance-readiness.pdf. Accessed March 12, 2022.

United Nations Environment Programme (UNEP), 2021. Adaptation Gap Report 2021. Retrieved from https://www.unep.org/resources/adaptation-gap-report-2021. Accessed November 28, 2021.

United Nations Framework Convention on Climate Change (UNFCCC), 2014. UNFCCC Standing Committee on Finance: 2014 Biennial Assessment and Overview of Climate Finance Flows Report. Retrieved from https://unfccc.int/files/cooperation_and_support/financial_mechanism/standing_committee/application/pdf/2014_biennial_assessment_and_overview_of_climate_finance_flows_report_web.pdf. Accessed August 17, 2021.

United Nations Framework Convention on Climate Change (UNFCCC), 2016. Key aspects of the Paris Agreement. Retrieved from https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement/key-aspects-of-the-paris-agreement. Accessed August 17, 2021.

United Nations Framework Convention on Climate Change (UNFCCC), 2021. *Adaptation Fund*. Retrieved from https://unfccc.int/Adaptation-Fund. Accessed August 17, 2021.

United Nations Framework Convention on Climate Change (UNFCCC), 2021. Matters relating to finance: New collective quantified goal on climate finance. Retrieved from https://unfccc.int/sites/default/files/resource/cma2021_L17E.pdf. Accessed March 13, 2022.

United Nations, 1992. Agenda 21. Retrieved from https://sustainabledevelopment.un.org/content/documents/Agenda21.pdf. Accessed August 17, 2021.

United Nations, 2021. Least Developed Countries Fund (LDCF). Retrieved from https://www.un.org/ldcportal/least-developed-countries-fund-ldcf/. Accessed November 28, 2021.

United Nations, 2021. Experts support UN Secretary-General's call for major effort in 2021 to achieve the $100 billion climate finance goal: New report recommends key actions to ramp up climate financing to developing countries. Retrieved from: https://www.un.org/sites/un2.un.org/files/finance_goal_report_release.pdf. Accessed March 13, 2022.

Watson, C., & Schalatek, L., 2020. Climate Finance Thematic Briefing: Adaptation Finance. Retrieved from https://climatefundsupdate.org/about-climate-finance/climate-finance-fundamentals/. Accessed March 2, 2021.

Watson, C., & Schalatek, L., 2020. Climate finance regional briefing: Small Island Developing States (Climate Finance Fundamentals No. 12). Retrieved from: https://climatefundsupdate.org/wp-content/uploads/2020/03/CFF12-2019-ENG-DIGITAL.pdf. Accessed March 2, 2021.

Watson, C., & Schalatek, L., 2021. The Global Climate Finance Architecture. Retrieved from https://climatefundsupdate.org/wp-content/uploads/2021/03/CFF2-ENG-2020-Digital.pdf. Accessed March 2, 2021.

Watson, C., Ballesteros, M., Buchner, B., Oliver, P., Ott, C., & Simon, G., 2019. 2018 Biennial Assessment and Overview of Climate Finance Flows: Technical Report. Retrieved from https://unfccc.int/sites/default/files/resource/2018%20BA%20Technical%20Report%20Final%20Feb%202019.pdf. Accessed August 17, 2021.

Weaver, A.J., Zickfeld, K., Montenegro, A., Eby, M., 2007. Long term climate implications of 2050 emission reduction targets. Geophys. Res. Lett. 34 (19) https://doi.org/10.1029/2007GL031018.

Weikmans, R., Roberts, J. T., & Robinson, S., 2020. 'What counts as climate finance? Define urgently. Nature, 588(7837), 220. https://www.nature.com/articles/d41586-020-03481-0.

Weikmans, R., Roberts, J.T., Baum, J., Bustos, M.C., Durand, A., 2017. Assessing the credibility of how climate adaptation aid projects are categorised. Dev. Pract. 27 (4), 458–471.

Weikmans, R., Asselt, H.V., Roberts, J.T., 2019. Transparency requirements under the Paris Agreement and their (un)likely impact on strengthening the ambition of nationally determined contributions (NDCs). Clim.ate Policy 20 (4), 511–526.

Weikmans, R., Roberts, J.T., 2019. The international climate finance accounting muddle: is there hope on the horizon? Clim. Dev. 11 (2), 97–111. https://doi.org/10.1080/17565529.2017.1410087.

Weiler, F., Klöck, C., Dornan, M., 2018. Vulnerability, good governance, or donor interests? The allocation of aid for climate change adaptation. World Dev. 104, 65–77. https://doi.org/10.1016/j.worlddev.2017.11.001.

World Bank. (2015). Finance for Climate Action: A snapshot of the World Bank Group's Climate Work. Retrieved from: https://www.worldbank.org/content/dam/Worldbank/document/Climate/FinanceClimateAction_Web.pdf. Accessed December 1, 2021.

World Bank, 2020. Climate Resilience: Championing Big Ambitions. Climate Investment Funds Factsheet. Retrieved from: https://www.climateinvestmentfunds.org/sites/cif_enc/files/ppcr_factsheet_web.pdf. Accessed August 17, 2021.

World Resources Institute (WRI), 2015. "Direct Access" to Climate Finance: Lessons Learned by National Institutions. Retrieved from: https://www.wri.org/research/direct-access-climate-finance-lessons-learned-national-institutions. Accessed March 12, 2022.
Yairen, J.C., 2022. Adapting to Climate Change Through Disaster Risk Reduction in the Caribbean: Lessons from the Global South in Tackling the Sustainable Development Goals. In: Flood, S., Columbié, Y.J., Tissier, M.L., O'Dwyer, B. (Eds.), Creating Resilient Futures: Integrating Disaster Risk Reduction, Sustainable Development Goals and Climate Change Adaptation Agendas. Palgrave Macmillan, Cham, pp. 183–203.
Yu, B. K., & Miller, S., 2011. Climate Change Funds and Implications for LAC Countries and the IDB. Inter-American Development Bank Technical Notes, No. IDB-TN-289.